K–Ar isochron age and paleomagnetism of diabase along the trans-Avalon aeromagnetic lineament—evidence of Late Triassic rifting in Newfoundland

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A prominent aeromagnetic lineament crosses the Avalon Peninsula of Newfoundland from 46°50.4′N, 53°45.9′W to 47°22.1′N, 52°30.0′W. It is shown to be at least partly caused by diabase dikes of Late Triassic and possibly Early Jurassic age which are probably related to the Shelburne diabase dike and the North Mountain basalt, both of Nova Scotia. All are thought to have resulted from rifting which preceded opening of the Atlantic.

Unmetamorphosed diabase was found at three sites along the trans-Avalon aeromagnetic lineament: as narrow sills at site 1 (46°58.0′N, 53°25.4′W), as a narrow dike at site 2 (47°4.7′N, 53°7.6′W), and as large angular boulders at site 3 (47°11.0′N, 52°52.2′W).

For sites 1 and 2, analyses of seven diabase samples fall on a single K–Ar isochron whose intercept on the $^{40}Ar/^{36}Ar$ axis is at 215 ± 45 and whose slope gives a Late Triassic age of 201.1 ± 2.6 Ma. Analyses of two diabase samples from the Shelburne dike fall close to this isochron suggesting a similar age. Paleomagnetism adds support; the virtual paleopole measured for sites 1 and 2, using 12 oriented diabase samples demagnetized in 300 Oe (23 880 A/m) alternating field (AF), falls at 87.8°E, 72.9°N (dp = 3.0°, dm = 4.3°), close to the virtual paleopole reported for the Shelburne dike.

For site 3, analyses of two samples fall on the K–Ar isochron reported for the North Mountain basalt, tentatively suggesting that the intrusion at site 3 occurred about 10 Ma later than at sites 1 and 2.

Un linéament aéromagnétique très marqué traverse la péninsule d'Avalon à Terre-Neuve de 46°50.4′N, 53°45.9′O jusqu'à 47°22.1′N, 52°30.0′O. On montre qu'il est causé au moins en partie par des dykes de diabase datant de la fin du Trias et peut-être du début du Jurassique, lesquels sont probablement reliés au dyke de diabase de Shelburne et au basalte de North Mountain, les deux en Nouvelle-Ecosse. On croit que toutes ces structures résultent d'un effondrement qui a précédé l'ouverture de l'Atlantique.

On a découvert de la diabase non métamorphisée à trois localités le long du linéament aéromagnétique trans-Avalon, dans des sills étroits au site 1 (46°58.0′N, 53°25.4′O), dans un dyke étroit au site 2 (47°4.7′N, 53°7.6′O) et dans de gros blocs anguleux au site 3 (47°11.0′N, 52°52.2′O).

Pour les sites 1 et 2, les analyses de sept échantillons de diabase tombent sur une même isochrone K–Ar dont l'intersection sur l'axe $^{40}Ar/^{36}Ar$ est de 215 ± 45 et dont la pente donne un âge de 201.1 ± 2.6 Ma correspondant à la fin du Trias. Les analyses de deux échantillons de diabase du dyke de Shelburne donnent des points qui se rapprochent de cette isochrone, ce qui laisse supposer un âge semblable. Les données de paléomagnétisme supportent cette hypothèse; le paléopôle virtuel mesuré pour les sites 1 et 2, en utilisant douze échantillons orientés de diabase désaimantés dans un champ alternatif de 300 Oe (23 880 A/m), se situe à 87.8°E, 72.9°N (dp = 3.0°, dm = 4.3°), ce qui se rapproche du pôle virtuel déterminé pour le dyke de Shelburne.

Pour le site 3, les analyses de deux échantillons se situent sur l'isochrone K–Ar rapportée pour le basalte de North Mountain, ce qui indiquerait que l'intrusion au site 3 s'est produite environ 10 Ma après celle des sites 1 et 2.

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Introduction

Papezik *et al.* (1975) drew attention to a prominent line of elongated aeromagnetic anomalies striking at about 60° across the Avalon Peninsula of Newfoundland (Fig. 1). They pointed out that the lineament is very likely due to a dike (or dikes) since it cuts across various rock types of late Precambrian age without noticeably displacing them. They suggested that this postulated dike might be an extension of the diabase dike north of Shelburne, Nova Scotia, since the two are in "nearly perfect alignment." That the two are part of a single dike now seems unlikely. There is only an approximate alignment when using lines of constant

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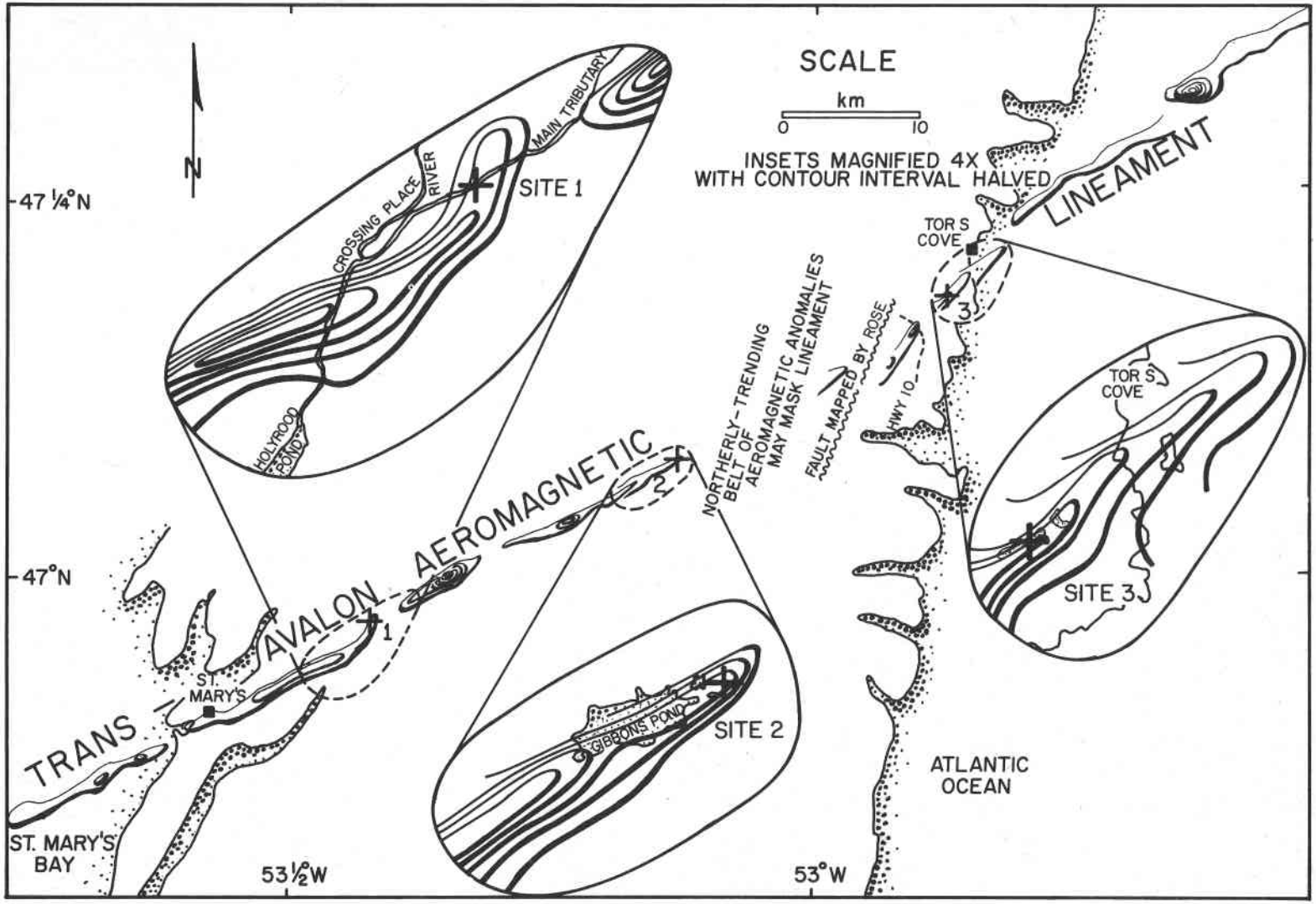


FIG. 1. Sampling sites on the aeromagnetic anomalies which define the trans-Avalon aeromagnetic lineament. The anomalies are drawn from Geological Survey of Canada (GSC) maps 7325G and 7329G with a 40 γ (40 nT) contour interval. The detailed insets are drawn from GSC maps 4538G, 4531G, and 4532G for sites 1, 2, and 3, respectively, with a 20 γ (20 nT) contour interval. Flight height was 1000 ft (305 m) above ground.

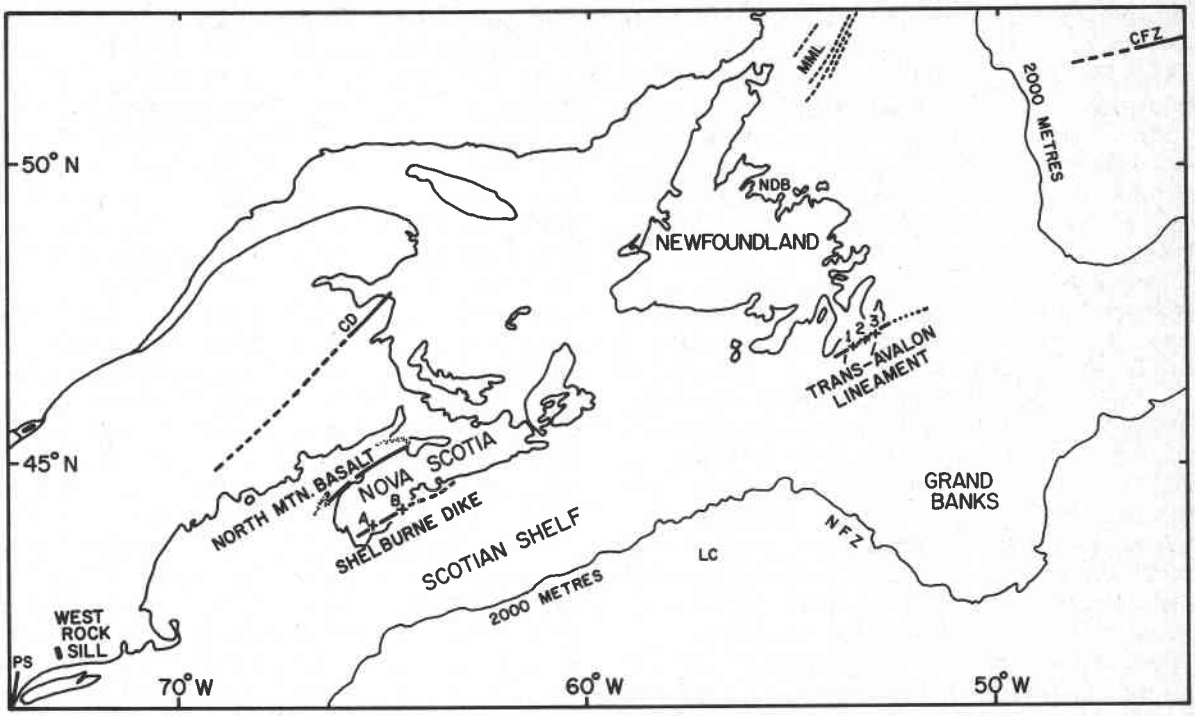


FIG. 2. Location of the trans-Avalon aeromagnetic lineament (with its marine magnetic extension dotted) relative to the Shelburne dike (with its offshore extension dashed). A Mercator projection is used since it has the unique property of showing all lines of constant bearing as straight lines. MML are marine magnetic lineaments (Haworth *et al.* 1976 *a*,*b*). CD is the Caraquet diabase dike with its aeromagnetic extension dashed (Burke *et al.* 1973). PS is the Palisade sill. NDB is Notre Dame Bay. LC is the Laurentian Cone. NFZ is the Newfoundland fracture zone complex. CFZ is the Charlie fracture zone.

bearing (Fig. 2) or great circles. Also, though Haworth traced the trans-Avalon lineament farther to the east (to 47°37′N, 51°33′W, R. T. Haworth, personal communication, 1977) using marine magnetic traverses, he could not trace it farther to the west. Besides, there may be dikes related to the Shelburne dike and parallel to it offshore on the Scotian Shelf (Hood 1967). Nevertheless, the basic suggestion—that the lineament may be related to the Shelburne dike, both resulting from Triassic–Jurassic rifting which preceded opening of the Atlantic—remained plausible and worth testing.

For the Shelburne dike, a Triassic K–Ar age (199 ± 32 Ma) and virtual paleopole had been reported (Larochelle and Wanless 1966). However, no dike had been found along the trans-Avalon lineament; even in recent mapping by Williams and King (1976), leading them to suggest that the lineament is due to "a basement feature beneath the [Hadrynian] Conception Group." This suggestion is ruled out because it requires the half width of the aeromagnetic anomaly to be of the order of the 4 km (Williams and King 1975) thickness of the Conception Group, whereas the half width is only of the order of the 0.3 km flight height. Indeed, the shape of ground magnetometer profiles shows that in places (e.g., at site 3 and where Highway 10 crosses the lineament) the postulated dikes should come to within 30 m of the surface. Hence, the postulated dikes must be younger than the Hadrynian rocks that the lineament cuts. How much younger was uncertain.

The approximate alignment of the trans-Avalon lineament and the Shelburne dike does suggest that the lineament is due to Triassic diabase dikes. Some support is added by borehole data which suggest that the initial rifting which preceded active spreading of the Atlantic began as early as the Late Triassic on the Grand Banks (Keen *et al.* 1977). There are less likely possibilities. The lineament might be due to Cambrian dikes associated with the opening of the "Proto-Atlantic" (discussed by Papezik *et al.* 1975). The lineament might be related to opening of the Atlantic between the Grand Banks and Galicia Bank though one would expect more northerly-striking dikes like the Late Jurassic to Early Cretaceous lamprophyres of central Newfoundland. The lineament might even be younger and related to marine magnetic lineaments to the northeast of Newfoundland (Fig. 2) which are

probably due to dikes that may cut strata as young as Cretaceous–Tertiary (Haworth *et al.* 1976*a*, *b*).

The aim of our present study was to find magnetic intrusives along the trans-Avalon lineament and to date them. We began with Hodych sampling unmetamorphosed diabase sills at site 1 which had been found by G. W. Einarson while helping Williams and King map the Trepassey area. Finding that the remanent magnetization of these sills yielded a paleopole close to that reported for the Shelburne dike, Hodych searched for more unmetamorphosed diabase along the lineament. A narrow diabase dike striking ~50° was found at site 2 and its remanent magnetization also yielded a paleopole close to that reported for the Shelburne dike. Large angular diabase boulders were found at site 3. Hayatsu then applied K–Ar isochron dating (Hayatsu 1972; Hayatsu and Carmichael 1977) to diabases from all three sites and from the Shelburne dike. K–Ar isochron dating proved itself more reliable than conventional K–Ar dating for these and related rocks (Table 3) because in each case the measured intercept of the isochron on the $^{40}Ar/^{36}Ar$ axis was significantly lower than the 295.5 assumed by conventional K–Ar dating.

Samples and Sites

The only strongly magnetic intrusions encountered along the trans-Avalon aeromagnetic lineament were unmetamorphosed diabases at sites 1, 2, and 3 and a large metamorphosed (greenschist facies) gabbro body sampled at 47°6.8′N, 53°1.9′W. The metamorphosed gabbro is younger than the Hadrynian Conception Group sediments that it intrudes but very likely older than the latest metamorphism known on the Avalon Peninsula which is Devonian. The metamorphosed gabbro is not likely related to the lineament since it is not confined to the vicinity of the lineament and similar magnetic gabbro bodies have been mapped extending northwards ~9 km (Mullins 1970 and aeromagnetic map 4531 G). Only the unmetamorphosed diabase will be discussed further.

At site 1 (46°58.0′N, 53°25.4′W), six oriented diabase blocks were collected with the aid of a sun compass—two blocks from each of three narrow sills (~0.3 m wide) a few metres apart. These sills should be representative of a number of narrow, northerly-striking, steeply-dipping diabase sills exposed nearby (within 0.5 km) in the bed of the main tributary of Crossing Place River (G. W. Einarson, H. Williams, and A. F. King, personal communication, 1975). The sills intrude nonmagnetic Hadrynian Conception Group sediments (Williams and King 1976). The sills seem to cause the northerly swing of the local termination of the trans-Avalon aeromagnetic lineament.

At site 2 (47°4.7′N, 53°7.6′W) six oriented diabase blocks were collected with the aid of a sun compass along the ~30 m exposed length of a narrow dike (~0.5 m wide). The dike intrudes a metamorphosed and much less magnetic gabbro stock (Mullins 1970 and ground magnetometry) which may be of Hadrynian age (McCartney 1967). The dike was noticed because of its roughly columnar jointing. The dike lies on the trans-Avalon aeromagnetic lineament near a local termination and has a strike of ~50° and a dip of ~80° southeast.

At site 3 (47°11.0′N, 52°52.2′W) no outcrop was found but there were a number of large angular blocks of magnetic diabase, two of which were sampled by drilling through their weathered exteriors. The diabase is coarser grained than at sites 1 and 2. The diabase blocks seem confined to the east end of a narrow pond that is aligned along the axis of the trans-Avalon lineament as located by ground magnetometry. The top of the dike (or dikes) at site 3 must be very shallow judging by the steepness of the flanks of the ground magnetometer anomaly. The only outcrops in the vicinity were of nonmagnetic Hadrynian sediments (St. John's Formation; Rose 1952) all of which lay to the side of the ground magnetometer anomaly.

The only other samples used were unoriented diabase blocks from the Shelburne dike—one block (FA650577) from Larochelle and Wanless' site A (43°50.3′N, 65°13.1′W) and one block (FA650587) from their site B (44°9.5′N, 64°29.0′W), both collected by W. F. Fahrig.

Paleopole Positions

One cylindrical specimen of 2.3 cm diameter and 2.1 cm length was studied from each of the 12 oriented blocks collected at sites 1 and 2. Remanences were measured on a Schonstedt SSM-1 magnetometer. Alternating field (AF) demagnetizations were done with three-axis unidirectional tumbling of the specimen.

The average direction of natural remanent magnetization (NRM) for each site is presented in Table 1 along with R (the vector sum of the unit vectors along each sample magnetization direction) and the usual measure of precision k and of dispersion α_{95} (Fisher 1953). The average intensity of NRM for site 1 is 3.41×10^{-3} emu/cm^3 and for site 2 is 7.94×10^{-3} emu/cm^3, the standard deviation being 1.06×10^{-3} emu/cm^3 and 2.30×10^{-3} emu/cm^3 respectively. The average magnetic susceptibility for site 1 is 3.12×10^{-3} emu/cm^3 and for site 2 is 2.21×10^{-3} emu/cm^3, the standard deviation being 0.30×10^{-3}

TABLE 1. Remanent magnetization directions

Sample locations	No. of samples	Laboratory treatment	Mean direction of remanence		Fisher statistics		
			Declination	Inclination	R	k	α_{95}
Site 1	6	NRM	7.2°	55.4°	5.897	48.6	9.7°
Site 2	6	NRM	11.3°	55.0°	5.947	93.9	7.0°
Site 1 + 2	12	NRM	9.2°	55.2°	11.841	69.3	5.3°
Site 1	6	300 Oe AF	14.6°	54.1°	5.982	278.5	4.0°
Site 2	6	300 Oe AF	11.0°	50.5°	5.968	156.1	5.4°
Site 1 + 2	12	300 Oe AF	12.7°	52.3°	11.942	190.2	3.1°

emu/cm³ and 0.39 × 10^{-3} emu/cm³ respectively. (The two Shelburne dike samples have an average NRM intensity of 0.77 × 10^{-3} emu/cm³ and an average susceptibility of 1.87 × 10^{-3} emu/cm³.)

All 12 specimens were AF demagnetized in 50 Oe (3980 A/m) steps from 100–500 Oe (7960–39 800 A/m), remeasuring after each step. On average, remanence intensities drop to 52.5, 14.0, 4.5, and 2.4% of NRM intensities after 100, 200, 300, and 400 Oe (7960, 15 920, 23 880, and 31 840 A/m) demagnetization, respectively, for site 1; and to 61.5, 28.5, 15.1, and 9.0% of NRM intensities for site 2. For both sites, the average direction of remanence moves a little farther from the present Earth's field direction as AF demagnetization progresses but seems to reach a stable end point and has minimum dispersion after 300 Oe (23 880 A/m). The results after 300 Oe (23 880 A/m) AF demagnetization are given in Table 1.

Larochelle and Wanless (1966) used the mean direction of their 11 samples after 350 Oe (27 860 A/m) AF demagnetization to calculate a virtual paleopole for sites A and B on the Shelburne dike. Similarly, we use the mean direction of our 12 specimens after 300 Oe (23 880 A/m) demagnetization to calculate a virtual paleopole for sites 1 and 2 on the trans-Avalon lineament. (These site 1 and 2 mean directions do pass Watson's (1956) F-ratio test showing that they are not significantly different at the 95% confidence level.) Our virtual paleopole is at 87.8°E, 72.9°N with dp = 3.0° and dm = 4.3° (where dp is the semi-axis of the ellipse of 95% confidence along the great circle through the site, and dm is the other semi-axis). This is only a virtual paleopole because the rocks did not record the field for the few thousands of years which may have been needed to average out secular variation. The ellipses of confidence for the trans-Avalon lineament and the Shelburne dike overlap showing that the two virtual paleopoles are not significantly different at the 95% confidence level (Fig. 3).

The magnetic properties of the two site 3 samples

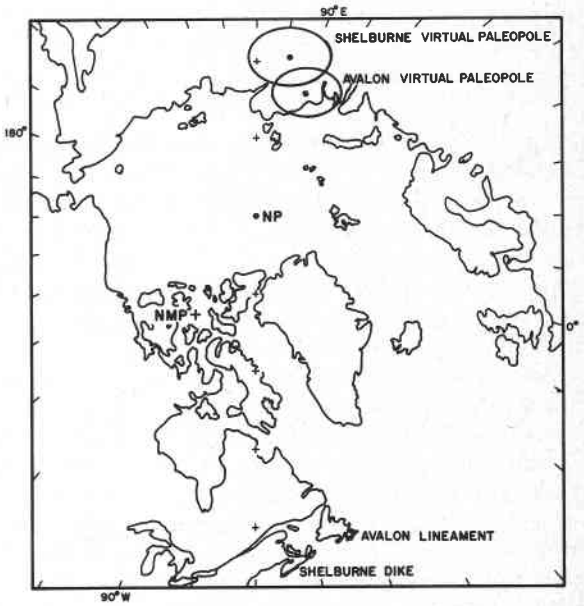


FIG. 3. Comparing the virtual paleopole position for trans-Avalon lineament sites 1 and 2, with that for Shelburne dike sites A and B, on a polar equidistant projection with ovals of 95% confidence.

resemble those of the site 1 and 2 samples. Their average intensity of NRM is 1.18 × 10^{-3} emu/cm³ and their average magnetic susceptibility is 2.52 × 10^{-3} emu/cm³. As the samples are AF demagnetized in 50 Oe (3980 A/m) steps from 100–250 Oe (7960–19 900 A/m), their remanence directions change by only a small amount; their remanence intensities drop less rapidly than for the site 1 and 2 samples.

Potassium–Argon Isochron Ages

In K–Ar isochron dating, it is important to remove virtually all atmospheric argon contamination from the samples before analysis, without removing any radiogenic argon. The procedure began with crushing the sample to pass 20 mesh, discarding powders finer than 100 mesh. A 5 g portion of crushed sample was immediately loaded into an

TABLE 2. K–Ar analytical data

Site	Sample No.	K (%)	$^{40}Ar_{rad.}$ ($\times 10^{-6}$ cm^3/g)	Conventional age (Ma)	$^{40}Ar_{total}$ ($\times 10^{-6}$ cm^3/g)	^{36}Ar ($\times 10^{-9}$ cm^3/g)	$^{40}Ar_{total}/^{36}Ar$	$^{40}K/^{36}Ar$ ($\times 10^5$)
1	2	0.3252	2.585	194	3.016	1.457	2070	1.493
1	4	0.2330	1.919	200	2.342	1.433	1634	1.088
1	6	0.2340	1.857	193	2.148	0.984	2181	1.590
2	1	0.2958	2.363	195	2.611	0.839	3113	2.360
2	3	0.3267	2.697	201	2.833	0.460	6155	4.750
2	4	0.3343	2.675	195	2.839	0.557	5098	4.017
2	5	0.2856	2.009	172	2.813	2.722	1034	0.702
3	3	0.7284	5.518	185	5.890	1.260	4673	3.867
3	4	0.7665	5.943	189	6.163	0.746	8264	6.877
A		0.8833	6.845	189	7.509	2.247	3342	2.631
B		0.6611	5.029	186	6.385	4.585	1392	0.964

NOTES: The constants used were: $\lambda_e = 0.581 \times 10^{-10}$/year; $\lambda_\beta = 4.962 \times 10^{-10}$/year; atomic $^{40}K/K = 1.167 \times 10^{-4}$.

argon extraction system (using a loading technique like technique (*a*) of Baksi (1973)). There the sample was baked at 190°C for 18 h in high vacuum. Hayatsu and Carmichael (1977) have shown that this baking removes atmospheric argon contamination as effectively as baking at 290°C for 60 h and that either baking leaves reproducibly the same amount of tightly-held ^{36}Ar and ^{40}Ar.

Baking was followed by argon analysis by the isotope dilution method, care being taken to maintain a minimal and reproducible system blank. The results corrected for system blank are listed in Table 2 along with a potassium analysis done on the remaining portion of the crushed sample (after further crushing it to pass 100 mesh). Measurements of potassium and radiogenic argon are reproducible to within 1%. Analytical errors of $^{40}Ar_{total}$ and ^{36}Ar are 1 or 2%. The ratio $^{40}Ar_{total}/^{36}Ar$ is plotted as a function of $^{40}K/^{36}Ar$ in Fig. 4. The uncertainty in "initial" ^{36}Ar may be several percent because of incomplete removal of genuine atmospheric contamination; but this should only shift the data points by several percent along lines connecting the data points to 295.5 on the $^{40}Ar/^{36}Ar$ axis, which should hardly affect the slope of the isochron.

Three samples from site 1 (one from each sill) and four samples from site 2 were analyzed. Within analytical error, all seven samples lie on a single isochron (Fig. 4) selected by least squares cubic fitting (York 1969). The isochron's slope yields an age of 201.1 Ma with a standard deviation of 2.6 Ma. The isochron's intercept on the $^{40}Ar/^{36}Ar$ axis is at 215 with a standard deviation of 45. Both slope and intercept are mostly defined by the four site 2 samples because the three site 1 samples have only a small range of $^{40}K/^{36}Ar$.

Two samples from site 3 were analyzed. Within analytical error, both lie on the isochron established for the North Mountain basalt of Nova Scotia (Hayatsu 1979).

The two samples from the Shelburne dike of Nova Scotia were also analyzed. The site A sample lies on the isochron for site 1 and 2 whereas the site B sample lies midway between this isochron and the North Mountain basalt isochron.

Discussion

Outcrops of the main dikes that likely cause the trans-Avalon aeromagnetic lineament have not been found. However, the unmetamorphosed diabases found at sites 1, 2, and 3 should represent the main dikes. The narrow diabase dike sampled at site 2 is very likely a minor extension of the main dikes since it lies on the lineament near a local termination and shares the lineament's strike (within ~10°). The narrow diabase sills sampled at site 1 have likely been fed from the main dikes causing the northward swing of the local termination of the lineament. Support is added by the site 1 and 2 diabases having the same paleopole and K–Ar isochron age suggesting that both sites were fed from a common source—very likely the main dikes. Even the angular boulders at site 3 may well represent a main dike since the boulders are on the lineament where ground magnetometry indicates that the top of the dike should be very shallow. Support is added by the site 3 diabases resembling the site 1 and 2 diabases in magnetic properties and K–Ar analyses—the latter suggesting that the intrusion at site 3 occurred ~10 Ma later than at sites 1 and 2. Presumably the main dikes are less resistant to erosion than most of the rocks they intrude; thus, except for narrow offshoots as at sites 1 and 2, they tend to occupy topographic depressions as at site 3.

The 201.1 ± 2.6 Ma K–Ar isochron age of the site 1 and 2 diabases should be their age of intrusion

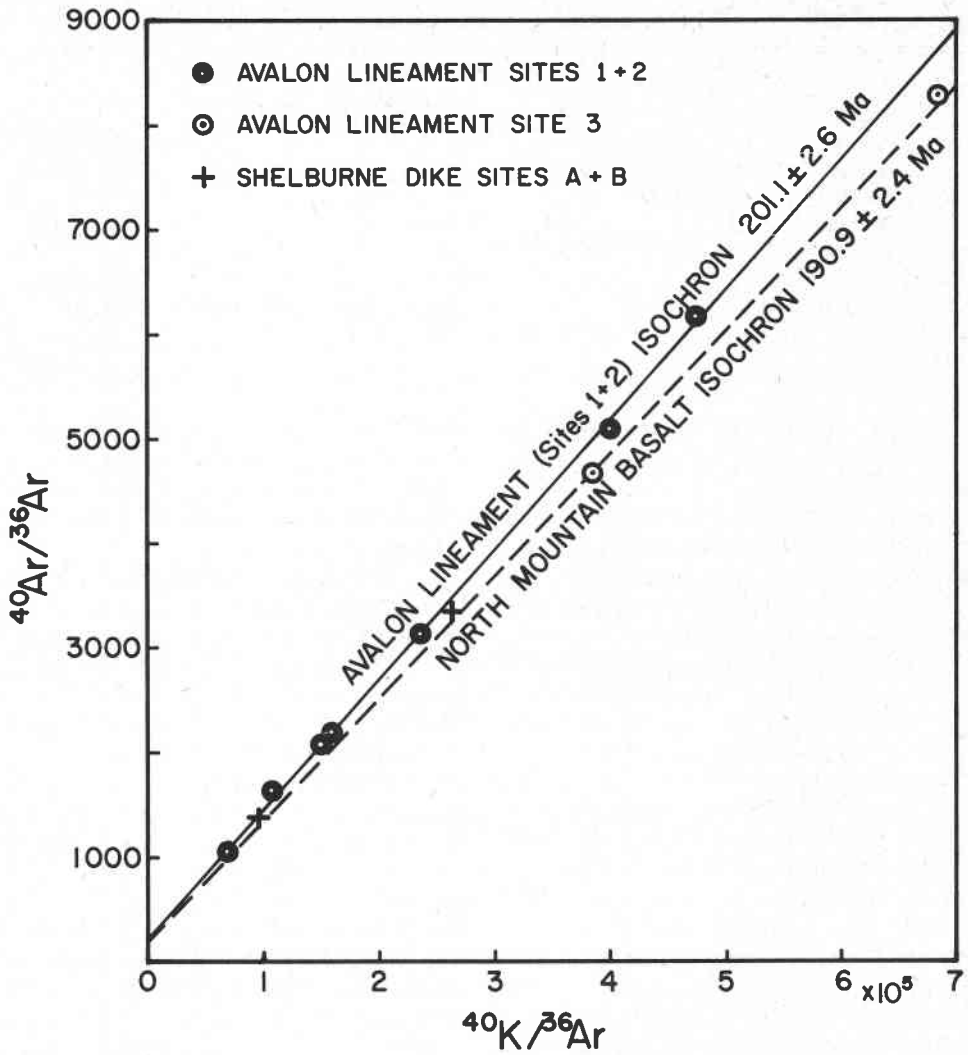


FIG. 4. Potassium–argon isochron plot for diabase samples from the trans-Avalon lineament sites 1, 2 (solid circles), and 3 (open circles) and from the Shelburne dike (crosses). The isochron for six North Mountain basalt samples (Hayatsu 1979) is shown dashed.

since no later geological events likely to have caused loss or gain of radiogenic argon are expected. The diabases along the trans-Avalon lineament are the only Triassic rocks for which we have evidence on the Island of Newfoundland. (There is palynological evidence for Triassic sediments offshore in wells on the Grand Banks (Barss *et al.* 1979).) The only younger rocks known on the island are dominantly northeast-trending Late Jurassic to Early Cretaceous lamprophyre dikes and a Late Jurassic gabbro stock, all in the Notre Dame Bay area of central Newfoundland (Strong and Harris 1974). The youngest metamorphic events known in the Avalon zone are Devonian (e.g., an $^{39}Ar/^{40}Ar$ study in progress by R. D. Dallmeyer on the Love Cove phyllites shows no evidence of metamorphism since the Devonian). Our diabases show no mineralogical evidence of even prehnite–pumpellyite facies metamorphism (according to a preliminary study by V. S. Papezik) and their weathering and alteration seem minor. That the conventional K–Ar ages for the site 1 and 2 samples (Table 2) range from the isochron age of 201 Ma down to 172 Ma is not evidence of argon loss; it is very likely due to the assumption of conventional K–Ar dating that nonradiogenic argon in the sample has $^{40}Ar/^{36}Ar$ equal to 295.5 whereas the measured intercept of the isochron is at 215 ± 45.

The two analyses for the Shelburne dike are not enough to define a K–Ar isochron age but suggest that the age of the Shelburne dike is close to that of the site 1 and 2 diabases. This is supported by the

TABLE 3. K–Ar isochron ages

Formation	No. of samples	K–Ar isochron age (Ma)	$^{40}Ar/^{36}Ar$ Tightly-bound nonradiogenic	Reference
Site 1 diabase sills and site 2 diabase dike trans-Avalon lineament, Newfoundland	7	201.1 ± 2.6	215 ± 45	This paper
Site 3 diabase dike? trans-Avalon lineament, Newfoundland	2 (not bedrock)	~191?		This paper
Shelburne diabase dike, Nova Scotia	2	~201?		This paper
North Mountain basalt, Nova Scotia	6	190.9 ± 2.4	195 ± 81	Hayatsu (1979)
West Rock diabase sill, Buttress diabase dike, Connecticut	11	190.6 ± 1.5	151 ± 55	Hayatsu (in preparation)

overlapping of the ovals of 95% confidence of the virtual paleopoles for the Shelburne dike and for the site 1 and 2 diabases.

The two analyses for site 3 diabases are to be used with caution because the samples were from boulders and may also be a little weathered. They were measured because there was some doubt as to whether the aeromagnetic lineaments east and west of site 2 were the same; the lineament cannot be traced through the belt of northerly-trending aeromagnetic anomalies to the east of site 2 (although this is not surprising) and is offset when it reappears (though this offset may be at least partly due to a high angle fault mapped by Rose (1952) and shown in Fig. 1). Both analyses for site 3 fall almost exactly onto the isochron for the North Mountain basalt; this is unlikely to be a fortuitous result of weathering and thus suggests that the site 3 diabases are of about the same age as the North Mountain basalt (190.9 ± 2.4 Ma, Hayatsu 1979). The North Mountain basalt is thought to be Upper Triassic because of the presence of *Isaura ovata* (Lea) in the underlying and overlying sediments (Klein 1962) but may be early Jurassic (since 191 Ma is close to the K–Ar isochron age of the West Rock sill of Connecticut (Table 3) and to the $^{40}Ar/^{39}Ar$ age of the Palisade sill (Dallmeyer 1975), both of which are expected from structure (Sanders 1963) and paleomagnetism (de Boer 1968) to be of similar age to the Holyoke and Hampden volcanics of Connecticut; these volcanics are in sediments which palynology suggests are Early Jurassic (Cornet *et al.* 1973)).

Conclusion

All the K–Ar isochron dating done for the Late Triassic to possible Early Jurassic of the Northern Appalachians is summarized in Table 3. Each isochron intercepts the $^{40}Ar/^{36}Ar$ axis at a point significantly lower than the 295.5 assumed by conventional K–Ar dating. This suggests that $^{40}Ar/^{36}Ar$ in the source region was significantly lower than 295.5 which would tend to make conventional K–Ar dates from these and related rocks too low but should not much affect the reliability of K–Ar isochron dates.

K–Ar isochron dating suggests that there were two episodes of intrusion along the trans-Avalon aeromagnetic lineament: at 201.1 ± 2.6 Ma ago and at ~191 Ma ago. K–Ar isochron data and paleomagnetism suggest that the first intrusion was roughly synchronous with intrusion of the Shelburne dike of Nova Scotia. (Hayatsu plans to test this further with more analyses of Shelburne dike samples.) K–Ar isochron data suggest that the second intrusion was roughly synchronous with extrusion of the North Mountain basalt of Nova Scotia and with intrusion of the West Rock diabase sill and the Buttress diabase dike of Connecticut. (We plan to test this further with work on diabase outcrops recently found ~2 km from site 3.) Both episodes of intrusion were presumably associated with rifting which preceded active sea-floor spreading in the Atlantic (May 1971) and support the suggestion (Keen *et al.* 1977) that there was Late Triassic rifting on the Grand Banks.

In contrast to this rifting, which our data suggest was roughly synchronous in Nova Scotia and Newfoundland at ~201 Ma ago and at ~191 Ma ago, the active sea-floor spreading which followed began in the Jurassic (probably ~172 Ma ago, Barrett and Keen 1976) off the Scotian Shelf but was delayed until the Cretaceous (probably ~115 Ma ago, Keen *et al.* (1977)) off the southeast margin of the Grand Banks. This delay to the north was presumably made possible by shearing along the southwest margin of the Grand Banks and along southern Iberia. Our data suggest that this shearing and the sea-floor spreading off the Scotian Shelf were not advanced enough by ~191 Ma ago to have decoupled Africa from North America and Iberia. They should have been advanced enough soon

after, by ~172 Ma ago if we accept Barrett and Keen's (1976) estimate of the age of the continental slope magnetic anomaly (perhaps earlier if we accept Uchupi and Austin's (1979) suggestion, based on acoustic stratigraphy, that the Laurentian Cone is underlain by Early Jurassic evaporites presumably deposited during an early-drift phase in a shallow basin fed by sea water entering through the Newfoundland fracture zone complex).

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